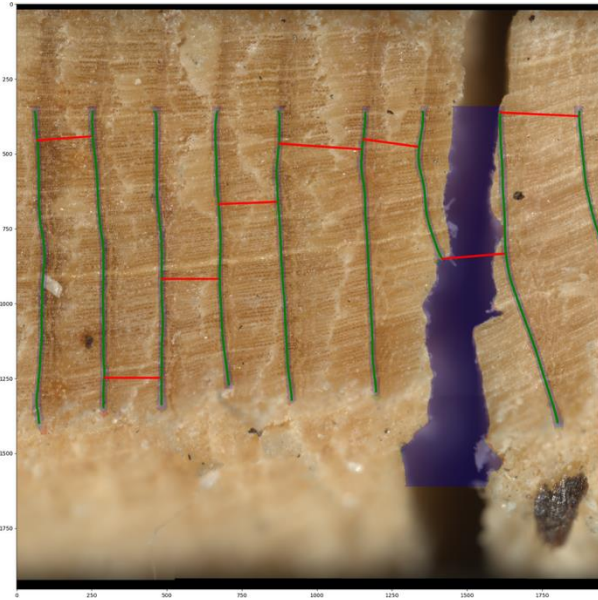


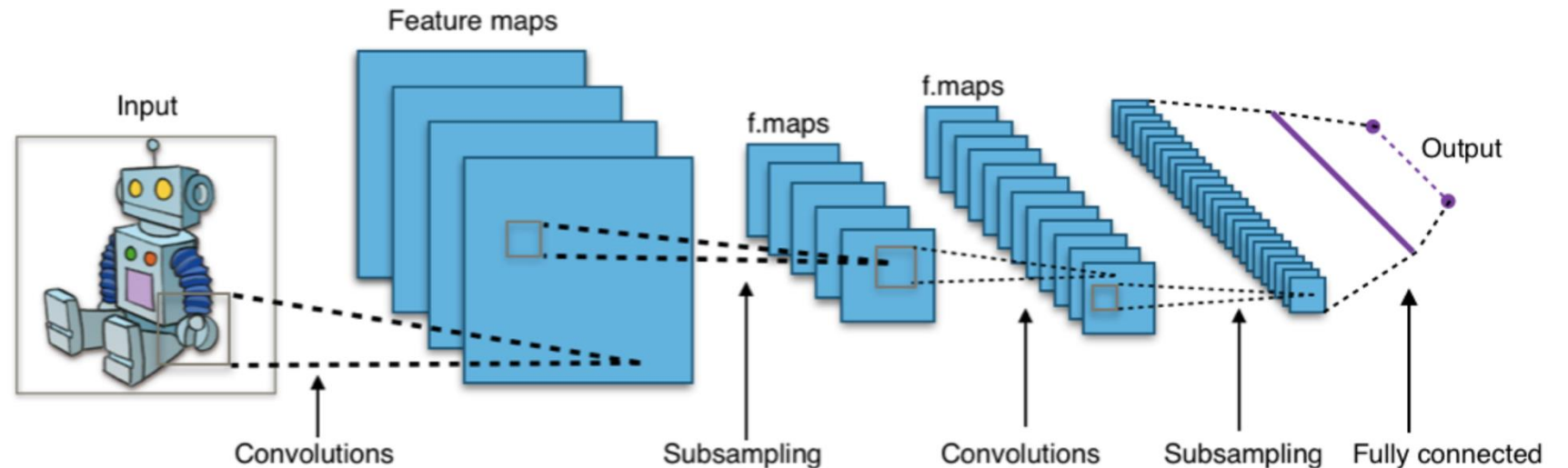
Automated ring-width measurements from images

Miroslav Poláček



Training neural networks

- Forward propagation
- Backward propagation – adjusting weights
- Multiple iterations - epochs



Training dataset design

- Important for generalisation
 - Should represent well the population
- Train, val split (80:20, 70:30)
 - Training – provides information to learn
 - Validation – used to choose what is a good weight
 - Test – we use real cores
- Data augmentation
 - Reduces overfitting
 - Improves generalisation

Can we use our current model on oak?

CVAT – open source annotation tool

- <https://app.cvat.ai>
- Preferably Google Chrome
- Create account if you did not
- New task
- Labels > Raw

```
[
  {
    "name": "RingBndy",
    "color": "#33ddff",
    "type": "polyline",
    "attributes": []
  },
  {
    "name": "Crack",
    "color": "#fa3253",
    "type": "polygon",
    "attributes": []
  }
]
```

CVAT – open source annotation tool

- Open the job
- Polyline for rings
- Polygon for cracks
- Use N to start and stop annotation
- Saving the annotations
 - Actions > Export task dataset > CVAT for images 1.1

Prepare dataset folder

- My_dataset
 - CVAT_annotations
 - annotations.xml
 - train
 - 2 oak image
 - val
 - 1 oak image

Arguments important for retraining

- **retraining_laptop.sh**

--training_data=../training/sample_dataset

- Triggers training process

--weights=../weights/best10px1000eAugEnlargedDataset.pt

--output_folder=../output

--run_ID=Retraining_run0

Arguments important for retraining

```
--epochs=1  
--annot_buffer=40  
--generate_annotations=True  
--debug=True
```

Arguments important for retraining

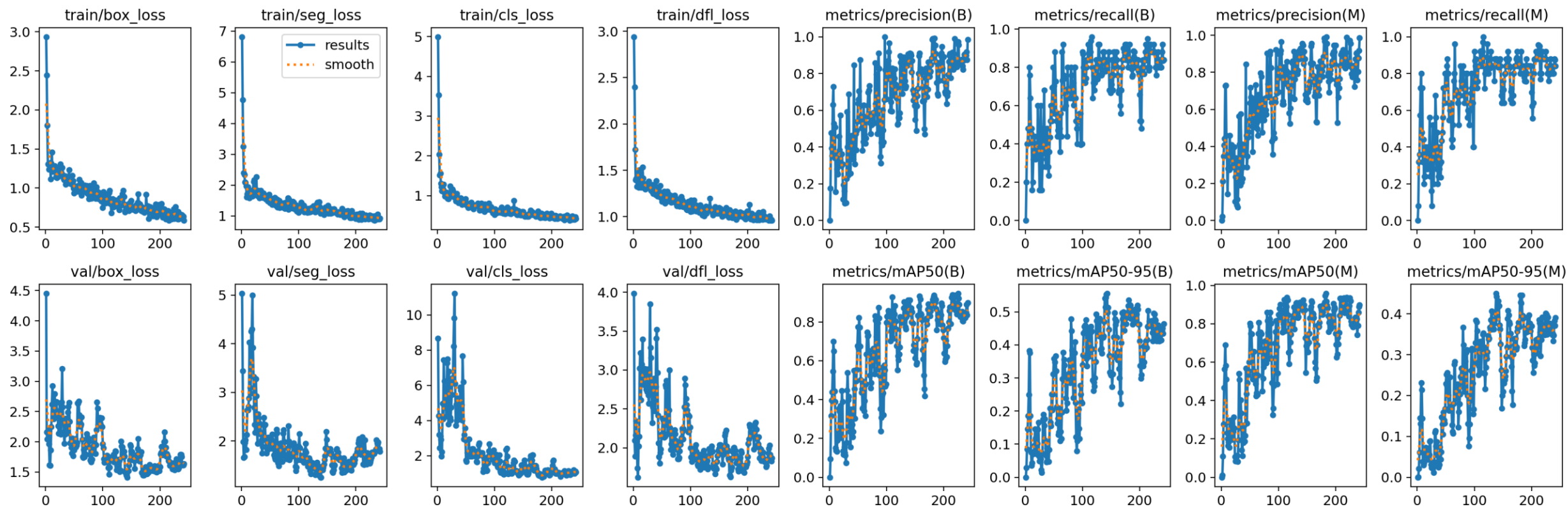
--generate_annotations=True
--debug=True

Let's run retraining

- **sh retraining_laptop.sh**
- On cluster
 - 2 cores, 34 train images (150 rings)
 - 1core, 16 val images (25 rings)

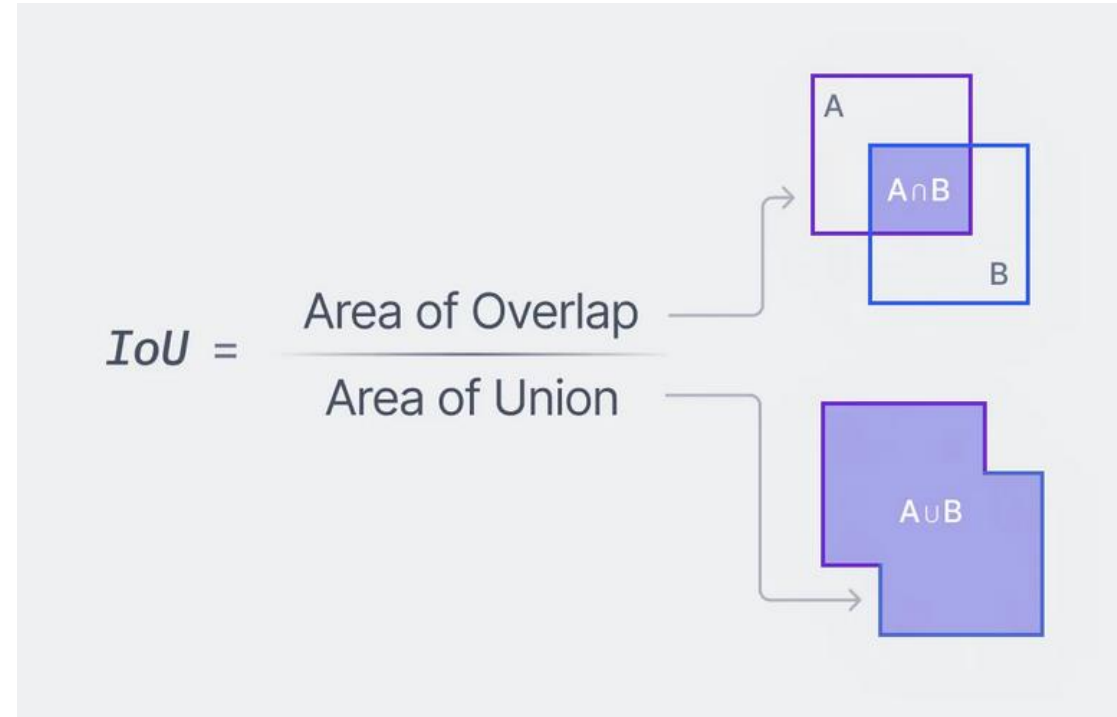
Check the results

results.png



Useful metrics

- **IoU**
- Precision
- Recall
- mAP



Useful metrics

- IoU
- **Precision**
- **Recall**

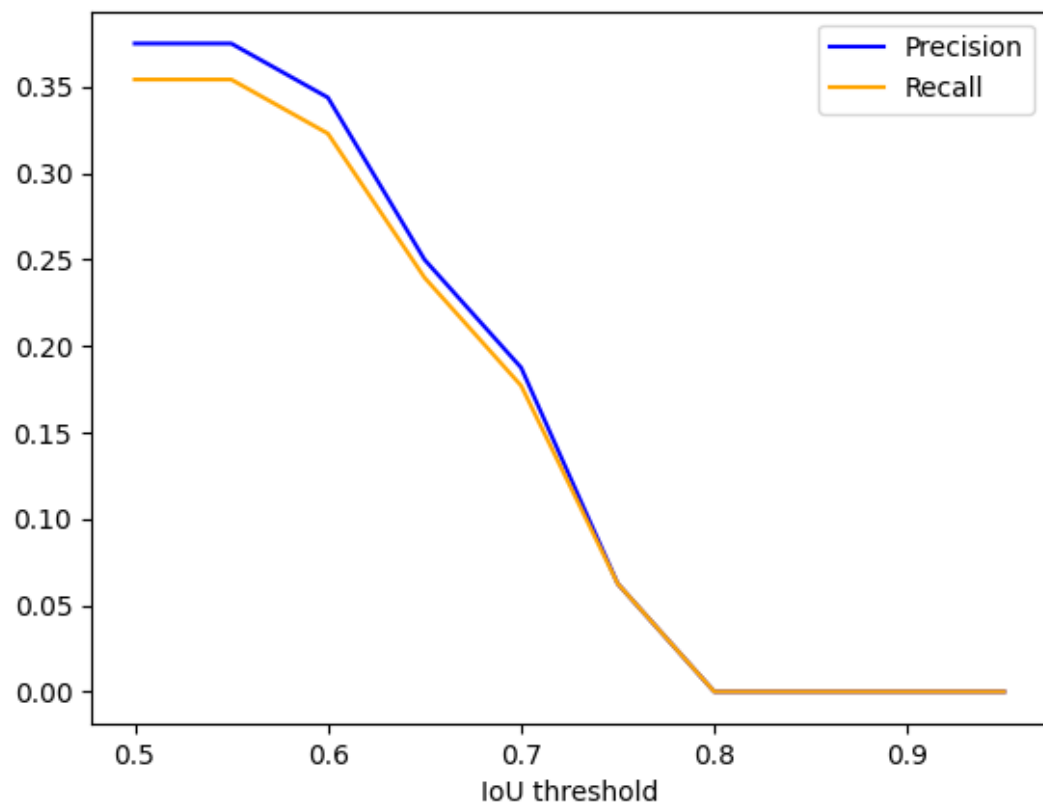
		Predicted	
		0	1
Actual	0	TN	FP
	1	FN	TP

$$Precision = \frac{TP}{TP + FP}$$

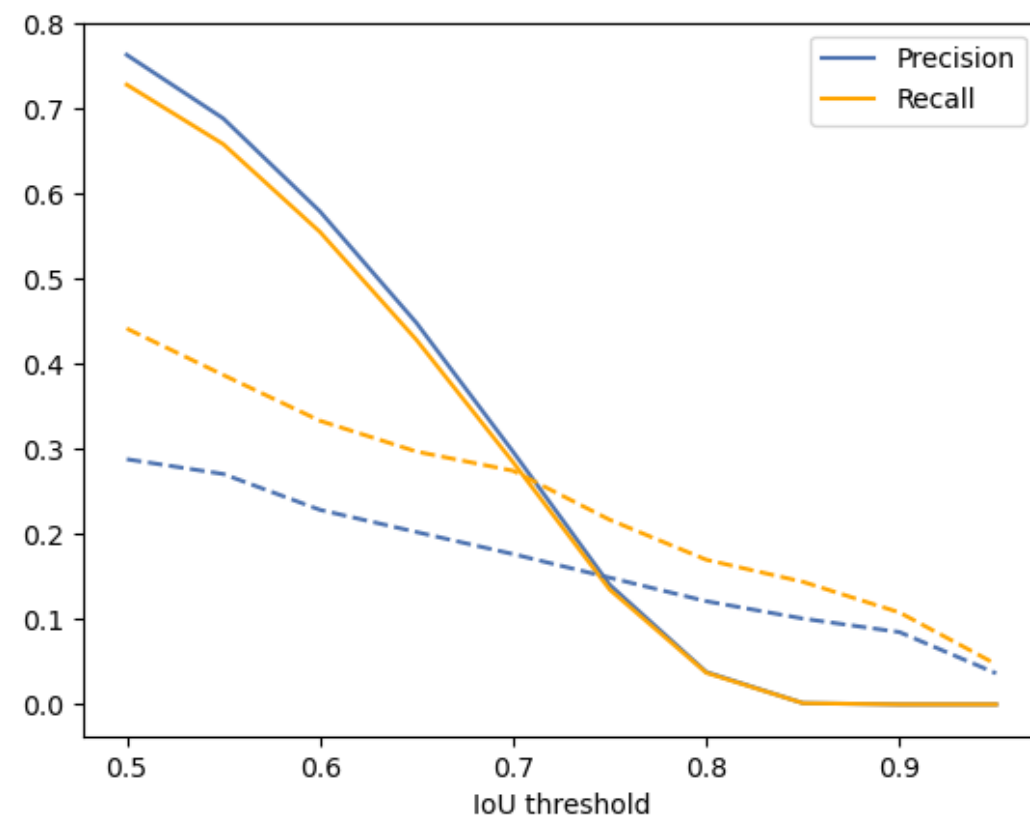
$$Recall = \frac{TP}{TP + FN}$$

Check results

Quercus



Current Spruce model



What can be done to improve the result?

What can be done to improve the result?

- More data
- Better data